```
## Assignment 1, Page 3: Access Corpora

from nltk.corpus import brown
from nltk.corpus import gutenberg
from nltk.corpus import webtext
from nltk.corpus import nps_chat
from nltk.corpus import reuters
from nltk.corpus import inaugural

#print(f'\nUsing words() on the Brown Corpus: {brown.words()}')
#print(f'\nUsing sents(): {brown.sents()}')
#print(f'\nUsing paras(): {brown.paras()}')
#print(f'\nUsing tagged_words(): {brown.tagged_words()}')
#print(f'\nUsing tagged_sents(): {brown.tagged_sents()}')
#print(f'\nUsing tagged_paras(): {brown.tagged_paras()}')
print(f'\nThe Inaugural Corpus broken up by word: {inaugural.words()}')
```

Access Corpora

Definitions: Corpus & Corpora

- The word *corpus* refers to one collection of texts.
- The word *corpora* refers to multiple collections of texts.
- We use the `nltk.corpus` package from the NLTK library to import corpora and apply corpus reader functions.

The corpora included in the NLTK library contain several genres of text.

Access Corpora

There are several corpora contained in the NLTK library. The most commonly used are listed and described below.

- **The Brown Corpus**: Created by Brown University in 1961, this was the first million-word electronic corpus of English.
- **The Gutenberg Corpus**: Containing 25,000 free electronic books, this data was taken from the Project Gutenberg electronic text archive of literature.
- **The Web Text Corpus**: Containing less formal language, this data was taken from an online discussion forum, personal advertisements and reviews, and so on.
- **The NPS Chat Corpus**: Originally created by the Naval Postgraduate School, this corpus contains over 10,000 posts from instant messaging chats.
- **The Reuters Corpus**: This corpus contains over 10,000 news documents, grouped into two sets called 'training' and 'test'.
- **The Inaugural Address Corpus**: This corpus contains each presidential inaugural address.

Enter the code below to access each of the above-mentioned corpora.

```
from nltk.corpus import brown
from nltk.corpus import gutenberg
from nltk.corpus import webtext
from nltk.corpus import nps_chat
from nltk.corpus import reuters
from nltk.corpus import inaugural
```

Use Corpus Reader Functions

We can use corpus reader functions to display corpus files in different arrangements. The most commonly used are listed and described below.

Enter the code for each reader function to see its functionality displayed, then select the `TRY IT` button to run the file.

.words(): Returns the contents of a specified corpus in a list of strings, where each string element in the list represents a word.

```
print(f'\nUsing words() on the Brown Corpus: {brown.words()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]
```

Your output should display as shown below.

```
g words() on the Brown Corpus: ['The', 'Fulton', 'County', 'Grand', 'Jury', 'said', ...
```

.sents(): Returns the contents of a specified corpus in a list of sub lists of strings, where each sub list is grouped by sentence and each string element in each sub list represents a word in the given sentence.

```
print(f'\nUsing sents(): {brown.sents()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]

Using sents(): [['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.'],
['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.'], ...]
```

Your output should display as shown below.

```
Using sents(): [['The', 'Fulton', 'County', 'Grand', 'Jury', 'said', 'Friday', 'an'
```

.paras(): Returns the contents of a specified corpus in a list of sub lists of sub lists of strings, where each sub list is grouped by paragraph, each sub list within that sub list is grouped by sentence, and each string element in each sub list of sub lists represents a word in the given sentence in the given paragraph.

```
print(f'\nUsing paras(): {brown.paras()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]

Using sents(): [['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.'],
['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.'], ...]

Using paras(): [[['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.']],
[['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.']], ...]
```

Your output should display as shown below.

```
Using paras(): [[['The', 'Fulton', 'County', 'Grand', 'Jury', 'said', 'Friday', 'an
```

.tagged_words(): Returns the contents of a specified corpus in a list of sub tuples of strings, where each sub tuple contains two string elements - the first string element being the word from the text and the second element being the given word's type of speech tag.

```
print(f'\nUsing tagged_words(): {brown.tagged_words()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]

Using sents(): [['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.'],
['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.'], ...]

Using paras(): [[['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.']],
[['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.']], ...]

Using tagged_words(): [('The', 'AT'), ('Fulton', 'NP-
TL'), ...]
```

Your output should display as shown below.

```
Using tagged_words(): [('The', 'AT'), ('Fulton', 'NP-TL'), ...]
```

.tagged.sents(): Returns the contents of a specified corpus in a list of sub lists of sub tuples of strings, where each sub list is grouped by sentence, and each sub tuple contains two string elements - the first string element being the word from the text and the second element being the given word's type of speech tag.

```
print(f'\nUsing tagged_sents(): {brown.tagged_sents()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]

Using sents(): [['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.'],
['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.'], ...]

Using paras(): [[['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.']],
[['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.']], ...]

Using tagged_words(): [('The', 'AT'), ('Fulton', 'NP-
TL'), ...]

Using tagged_sents(): [[('The', 'AT'), ('Fulton',
'NP-TL'), ('County', 'NN-TL'), ('Grand', 'JJ-TL'),
('Jury', 'NN-TL'), ('said', 'VBD'), ('Friday', 'NR'),
('an', 'AT'), ('investigation', 'NN'), ('of', 'IN'),
("Atlanta's", 'NP$'), ('recent', 'JJ'), ('primary',
'NN'), ('election', 'NN'), ('produced', 'VBD'),
('``', '``'), ('no', 'AT'), ('evidence', 'NN'),
("''", "''"), ('that', 'CS'), ('any', 'DTI'),
('irregularities', 'NNS'), ('took', 'VBD'), ('place',
'NN'), ('.', '.')], [('The', 'AT'), ('jury', 'NN'),
('further', 'RBR'), ('said', 'VBD'), ('in', 'IN'),
('term-end', 'NN'), ('presentments', 'NNS'), ('that',
'CS'), ('the', 'AT'), ('City', 'NN-TL'),
('Executive', 'JJ-TL'), ('Committee', 'NN-TL'), (',',
','), ('which', 'WDT'), ('had', 'HVD'), ('over-all',
'JJ'), ('charge', 'NN'), ('of', 'IN'), ('the', 'AT'),
('election', 'NN'), (',', ','), ('``', '``'),
('deserves', 'VBZ'), ('the', 'AT'), ('praise', 'NN'),
('and', 'CC'), ('thanks', 'NNS'), ('of', 'IN'),
('the', 'AT'), ('City', 'NN-TL'), ('of', 'IN-TL'),
('Atlanta', 'NP-TL'), ("''", "''"), ('for', 'IN'),
('the', 'AT'), ('manner', 'NN'), ('in', 'IN'),
('which', 'WDT'), ('the', 'AT'), ('election', 'NN'),
('was', 'BEDZ'), ('conducted', 'VBN'), ('.', '.')],
...]
```

Your output should display as shown below.

```
Using tagged_sents(): [[('The', 'AT'), ('Fulton', 'NP-TL'), ('County', 'NN-TL'), ('
```

.tagged_paras(): Returns the contents of a specified corpus in a list of sub lists of sub lists of sub tuples of strings, where each sub list is grouped by paragraph, each sub list within a given sub list is grouped by sentence, and each sub tuple contains two string elements - the first string element being the word from the text and the second element being the given word's type of speech tag.

```
print(f'\nUsing tagged_paras(): {brown.tagged_paras()}')
```

TRY IT

```
Using words() on the Brown Corpus: ['The', 'Fulton',
'County', 'Grand', 'Jury', 'said', ...]

Using sents(): [['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.'],
['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.'], ...]

Using paras(): [[['The', 'Fulton', 'County', 'Grand',
'Jury', 'said', 'Friday', 'an', 'investigation',
'of', "Atlanta's", 'recent', 'primary', 'election',
'produced', '``', 'no', 'evidence', "''", 'that',
'any', 'irregularities', 'took', 'place', '.']],
[['The', 'jury', 'further', 'said', 'in', 'term-end',
'presentments', 'that', 'the', 'City', 'Executive',
'Committee', ',', 'which', 'had', 'over-all',
'charge', 'of', 'the', 'election', ',', '``',
'deserves', 'the', 'praise', 'and', 'thanks', 'of',
'the', 'City', 'of', 'Atlanta', "''", 'for', 'the',
'manner', 'in', 'which', 'the', 'election', 'was',
'conducted', '.']], ...]

Using tagged_words(): [('The', 'AT'), ('Fulton', 'NP-
TL'), ...]

Using tagged_sents(): [[('The', 'AT'), ('Fulton',
'NP-TL'), ('County', 'NN-TL'), ('Grand', 'JJ-TL'),
('Jury', 'NN-TL'), ('said', 'VBD'), ('Friday', 'NR'),
('an', 'AT'), ('investigation', 'NN'), ('of', 'IN'),
("Atlanta's", 'NP$'), ('recent', 'JJ'), ('primary',
'NN'), ('election', 'NN'), ('produced', 'VBD'),
('``', '``'), ('no', 'AT'), ('evidence', 'NN'),
("''", "''"), ('that', 'CS'), ('any', 'DTI'),
('irregularities', 'NNS'), ('took', 'VBD'), ('place',
'NN'), ('.', '.')], [('The', 'AT'), ('jury', 'NN'),
('further', 'RBR'), ('said', 'VBD'), ('in', 'IN'),
('term-end', 'NN'), ('presentments', 'NNS'), ('that',
'CS'), ('the', 'AT'), ('City', 'NN-TL'),
('Executive', 'JJ-TL'), ('Committee', 'NN-TL'), (',',
','), ('which', 'WDT'), ('had', 'HVD'), ('over-all',
'JJ'), ('charge', 'NN'), ('of', 'IN'), ('the', 'AT'),
('election', 'NN'), (',', ','), ('``', '``'),
('deserves', 'VBZ'), ('the', 'AT'), ('praise', 'NN'),
('and', 'CC'), ('thanks', 'NNS'), ('of', 'IN'),
('the', 'AT'), ('City', 'NN-TL'), ('of', 'IN-TL'),
('Atlanta', 'NP-TL'), ("''", "''"), ('for', 'IN'),
('the', 'AT'), ('manner', 'NN'), ('in', 'IN'),
('which', 'WDT'), ('the', 'AT'), ('election', 'NN'),
('was', 'BEDZ'), ('conducted', 'VBN'), ('.', '.')],
...]

Using tagged_paras(): [[[('The', 'AT'), ('Fulton',
'NP-TL'), ('County', 'NN-TL'), ('Grand', 'JJ-TL'),
('Jury', 'NN-TL'), ('said', 'VBD'), ('Friday', 'NR'),
('an', 'AT'), ('investigation', 'NN'), ('of', 'IN'),
("Atlanta's", 'NP$'), ('recent', 'JJ'), ('primary',
'NN'), ('election', 'NN'), ('produced', 'VBD'),
('``', '``'), ('no', 'AT'), ('evidence', 'NN'),
("''", "''"), ('that', 'CS'), ('any', 'DTI'),
('irregularities', 'NNS'), ('took', 'VBD'), ('place',
'NN'), ('.', '.')]], [[('The', 'AT'), ('jury', 'NN'),
('further', 'RBR'), ('said', 'VBD'), ('in', 'IN'),
('term-end', 'NN'), ('presentments', 'NNS'), ('that',
'CS'), ('the', 'AT'), ('City', 'NN-TL'),
('Executive', 'JJ-TL'), ('Committee', 'NN-TL'), (',',
','), ('which', 'WDT'), ('had', 'HVD'), ('over-all',
'JJ'), ('charge', 'NN'), ('of', 'IN'), ('the', 'AT'),
('election', 'NN'), (',', ','), ('``', '``'),
('deserves', 'VBZ'), ('the', 'AT'), ('praise', 'NN'),
('and', 'CC'), ('thanks', 'NNS'), ('of', 'IN'),
('the', 'AT'), ('City', 'NN-TL'), ('of', 'IN-TL'),
('Atlanta', 'NP-TL'), ("''", "''"), ('for', 'IN'),
('the', 'AT'), ('manner', 'NN'), ('in', 'IN'),
('which', 'WDT'), ('the', 'AT'), ('election', 'NN'),
('was', 'BEDZ'), ('conducted', 'VBN'), ('.', '.')]],
...]
```

Your output should display as shown below.

```
Using tagged_paras(): [[[('The', 'AT'), ('Fulton', 'NP-TL'), ('County', 'NN-TL'), (
```

Try this variation:

Write the code to produce the following expected output, then select the `TRY IT` button to run the file and check your work.

The expected output is shown below.

```
The Inaugural Corpus broken up by word: ['Fellow', '-', 'Citizens',
```

TRY IT

```
The Inaugural Corpus broken up by word:
['Fellow', '-', 'Citizens', 'of', 'the',
'Senate', ...]
```

▶ **Solution**

Reading Question

Fill in the blanks.

Statement	**Solution**
We can access various corpora by using the ? command, which is followed by the `nltk.corpus` command.	from
This is followed by the ? command, which is followed by the name of the corpus.	import
To use a reader function, such as `sents`, we type the ? of the corpus.	name
This is followed by a ?, which is followed by the name of the reader function, followed by parentheses.	period

We can access various corpora by using the **from** command followed by the command *nltk.corpus* followed by the **import** command followed by the name of the corpus. An example of this is shown below.

```
from nltk.corpus import gutenberg
```

To use a reader function, such as *.sents()*, we type the **name** of the corpus, followed by a **period** followed by the name of the reader function, followed by parentheses. An example of this is shown below.

```
gutenberg.sents()
```

Check It! (1 left)

Next